



4.4 Probability, Venn and tree diagrams

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For events A and B: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (addition rule); if mutually exclusive, $P(A \cap B) = 0$. Conditional probability $P(A|B) = P(A \cap B)/P(B)$. Events A and B are independent if $P(A \cap B) = P(A) \times P(B)$, equivalently $P(A|B) = P(A)$. Venn diagrams display overlapping sets; tree diagrams display sequential/conditional probabilities, with branch probabilities multiplied along a path and added across paths for a given outcome.

Worked example: follow the reasoning

In a class, $P(\text{studies French}) = 0.5$, $P(\text{studies Spanish}) = 0.4$, $P(\text{studies both}) = 0.2$. Find $P(\text{studies French or Spanish})$.

$$P(F \cup S) = P(F) + P(S) - P(F \cap S) = 0.5 + 0.4 - 0.2 = 0.7.$$

Quick check · multiple choice

Two events A and B are independent with $P(A) = 0.3$ and $P(B) = 0.5$. Find $P(A \cap B)$.

- A 0.15
- B 0.8
- C 0.2
- D 0.35

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A bag contains 5 red and 3 blue balls. Two balls are drawn at random without replacement. [8]
- (a) Draw/describe a tree diagram showing the probabilities for the colours of the two balls drawn.
 - (b) Find the probability that both balls are the same colour.
 - (c) Given that the two balls drawn are the same colour, find the probability that both are red.

Final answer: _____



Answers and explanation

Question 1 [8]

- (a) First draw: $P(R)=5/8$, $P(B)=3/8$. Second draw (without replacement): if first is R, $P(R|R)=4/7$, $P(B|R)=3/7$; if first is B, $P(R|B)=5/7$, $P(B|B)=2/7$.
- (b) $P(\text{same colour}) = P(RR)+P(BB) = (5/8)(4/7) + (3/8)(2/7) = 20/56 + 6/56 = 26/56 = 13/28$.
- (c) $P(RR|\text{same}) = P(RR)/P(\text{same}) = (20/56)/(26/56) = 20/26 = 10/13$.

Quick check answer

A. For independent events, $P(A \cap B) = P(A) \times P(B) = 0.3 \times 0.5 = 0.15$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.